3 Computational thinking: Making a good plan

You will learn

- how to plan a program
- what an algorithm is
- what happens when you run a program.

In this unit you will learn to make and use a plan. A plan sets out the steps you have to take to solve a problem. Good planning can help you to achieve your goals.



Class activity

Another name for the plan for a task is an algorithm.

- What is your favourite food? Do you know how to make that food? Write down the actions in the right order.
- What is your favourite computer game? Write down what happens in the game. Put the actions in the right order.

These are examples of algorithms. Make an algorithm into a poster for the classroom wall.



Learning outcomes: Say what an algorithm is and what running a program means

Talk about...

Share the recipe plans and game plans you made. Work together to make the plans even better. Can you add more actions? Can you add pictures?

algorithm
depends on
run a program
command



Did you know?

On internet sites, such as YouTube, you can watch videos of people cooking their favourite foods.



4.1 Choose a sprite

In this lesson

You will learn:

- how to choose a sprite for your program.

Spiral back



Last year you used a simple program. It controlled a sprite. Now you will write a program to control a sprite.

In Unit 3 you played the Frog Hop Game. In this unit you will make a simple version of the game. You can choose any sprite you like. It does not have to be a frog.

You will make a Scratch program. Start on the Scratch website.

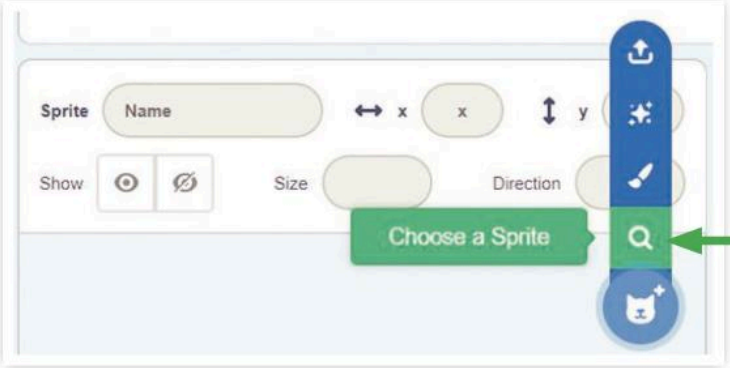
Choose a sprite

Look at the bottom right. This part of the screen shows the sprites.



Sprite 1 is a cat. Delete this sprite by clicking on the cross.

You will choose a new sprite.



Click here to see the other sprites.

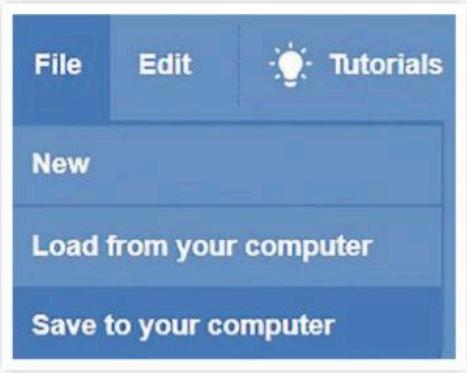
You will see pictures of sprites.



Click on the one you choose.

Save program file

Now you must save your work. You will save it on your computer. Open the File menu and choose 'Save to your computer'. Then type a name for your file.





Activity

Use the skills you learned to:

- pick a sprite
- save your file.

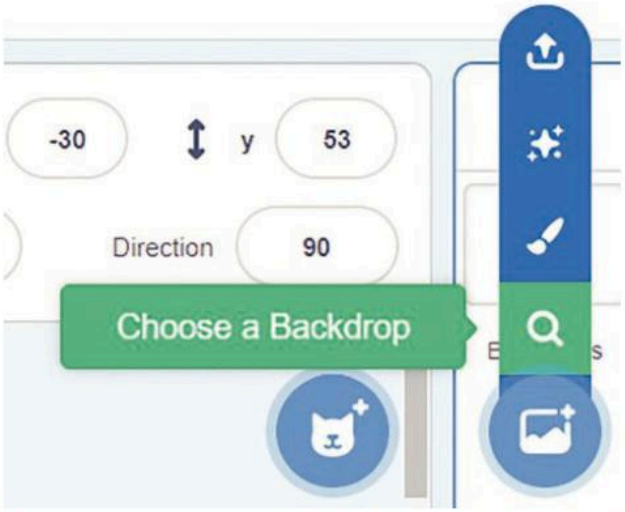


Draw the sprite you chose. Imagine a computer game with your sprite in it. Write down your idea.



Extra challenge

Add a backdrop to your program.



4.2 Command your sprite

In this lesson

You will learn:

➔ how to add commands to your program.

Load program file

Last lesson you saved your work as a file. Now you will load the file. Load means get a file from storage.



Open the file menu. Click on 'load from your computer'. Choose the file you made last time.

Coloured dots and blocks

You will make a Scratch program. A program is made of **commands**. Commands tell the computer what to do.

Scratch commands are blocks. You fit the blocks together. That makes the program.

Different types of blocks are different colours.

- 'Motion' blocks are blue.
- 'Sound' blocks are pink.

Look for the coloured dots on the left of the screen. Choose a colour to help you pick the blocks you need.

Actions

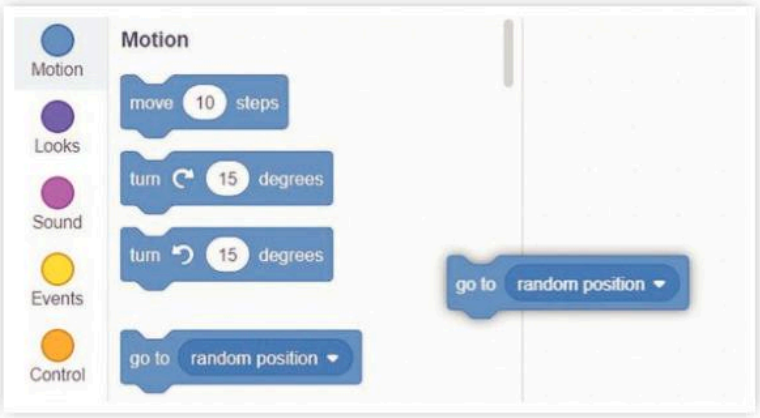
Choose a block. It will make your sprite move.

Click on the blue dot. You will see 'Motion' blocks.

Find the block that says 'go to random position'.

Drag this block to the middle of the screen.

Click on the block. The sprite will move.



Sounds

Choose a block. It will play a sound. Click on the pink dot. You will see 'Sound' blocks. Choose the 'play sound' block.

Now you have two blocks. Fit the blocks together.

Click on the two blocks. What happens?





Activity

Make the two-block program shown on this page.



Extra challenge

Add any other block to the program. See what happens when you run the program.

Think again

Say what blocks you used. Say what each block did.

4.3 A start event

In this lesson

You will learn:

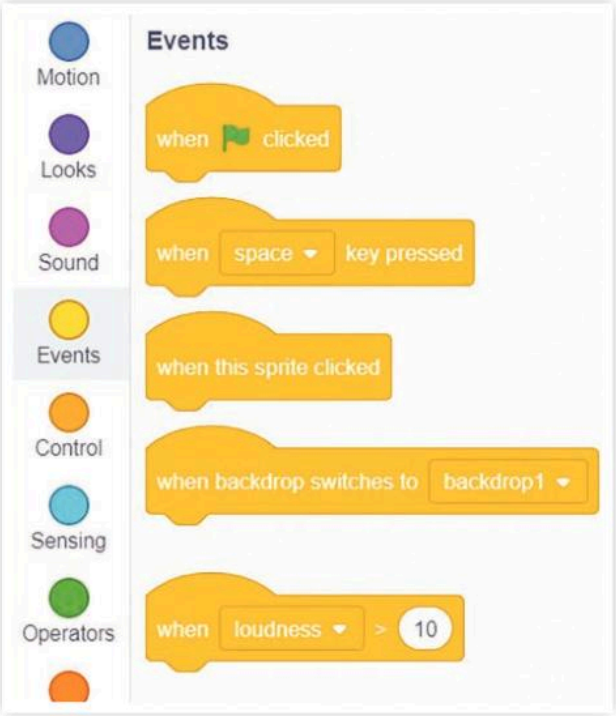
- to choose an event that starts the program.

Events

An **event** means anything that happens. In Scratch 'Events' are things the **user** does. The user is the person who uses the program. Now you will choose the 'Event' that lets the user start your program.

Click on the yellow dot that says 'Events'.

What 'Event' blocks can you see? What does each one say?

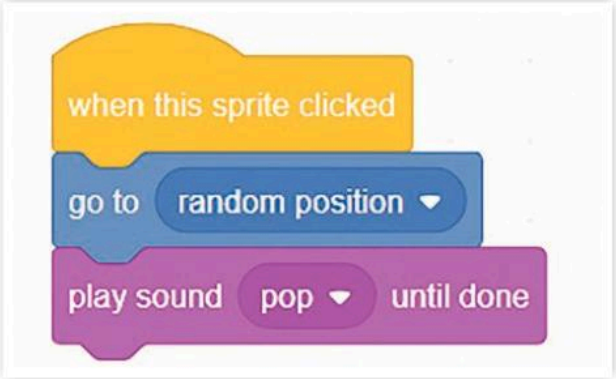


Choose an event

To choose an 'Event' drag the block to the program area.

Find the 'Event' block that says 'when this sprite clicked'.

Fit the program blocks onto the 'Event' block. Now the event will start the program.



Change colour

Now you will make the sprite change colour. Click on the purple dot that says 'Looks'. These command blocks change the look of the sprite.

Find the block that says 'change color effect by…'.

Drag this block to the script area and add it to the program.





Activity

Change your program.

- Add the 'Event' block which says 'when this sprite clicked'.
- Add the 'Looks' block which changes the colour of the sprite.

Click the sprite to make it move.

Save the file.



Extra challenge

Choose a different start event. Change the program to the new event.

Think again

Draw the 'Event' block you used. Say what it does.

If you used more than one, draw them all.